Experiment-8

Aim : To study and implement **GROUP BY** and **ORDER BY** clauses using a *Sales* table.

- ✓ **GROUP BY** → Used to group rows with same values and perform aggregate functions like SUM, AVG.
- ✓ **ORDER BY** → Used to sort the result in ascending or descending order.

Procedure & Queries

- ✓ Create database Lab_Assignment_8 and Use database.

```
CREATE DATABASE Lab_Assignment_8;
▷Run
USE Lab_Assignment_8;
```

Step 1: Create Sales Table

```
CREATE TABLE Sales (
    SalesNo INT,
    SalesName VARCHAR(50),
    Branch VARCHAR(50),
    SalesAmount DECIMAL(10,2),
    DOB DATE
);
```

Explanation

- A table named **Sales** is created.
- It stores salesman details like name, branch, sales amount, and date of birth.

Step 2: (a) Insert Five Records

```
INSERT INTO Sales VALUES
(1, 'Jony Gautam', 'Delhi', 50000, '2000-12-21'),
(2, 'Varun', 'Mumbai', 60000, '1999-05-10'),
(3, 'Pranjal', 'Delhi', 55000, '2001-12-15'),
(4, 'Anushka Mishra', 'Chennai', 45000, '1998-07-25'),
(5, 'Ankit', 'Mumbai', 70000, '2000-12-05');
```

Explanation

- Five records are inserted into the Sales table.
- Each record contains branch and sales details.

(b) Calculate Total Sales Amount in Each Branch

```
SELECT Branch, SUM(SalesAmount) AS TotalSales
FROM Sales
GROUP BY Branch;  6ms
```

Output:

Branch varchar	TotalSales decimal
Delhi	105000.00
Mumbai	130000.00
Chennai	45000.00

Explanation

- Groups records by **Branch**
- Calculates **total sales** using SUM()

(c) Calculate Average Sales Amount in Each Branch

```
SELECT Branch, AVG(SalesAmount) AS AvgSales
FROM Sales
GROUP BY Branch;  6ms
```

Output:

Branch varchar	AvgSales decimal
Delhi	52500.00000(
Mumbai	65000.00000(
Chennai	45000.00000(

Explanation

- Groups data by branch
- Calculates **average sales** using AVG()

(d) Display Salesmen Born in December (Format: 21-Dec-09)

```
SELECT SalesName,
DATE_FORMAT(DOB, '%d-%b-%y') AS FormattedDOB
FROM Sales
WHERE MONTH(DOB) = 12;  6ms
```

Output:

SalesName varchar	FormattedDOB varchar
Jony Gautam	21-Dec-00
Pranjal	15-Dec-01
Ankit	05-Dec-00

Explanation

- MONTH(DOB) = 12 → Filters December births
- DATE_FORMAT() → Converts date into format like **21-Dec-09**

(e) Display Name and DOB in Alphabetical Order of Month

```
SELECT SalesName, DOB
FROM Sales
ORDER BY MONTHNAME(DOB);   16ms
```

Output:

SalesName varchar	DOB date
Jony Gautam	2000-12-21
Pranjal	2001-12-15
Ankit	2000-12-05
Anushka Mishra	1998-07-25
Varun	1999-05-10

Explanation

- MONTHNAME(DOB) returns month name (January, February...)
- ORDER BY sorts them **alphabetically by month name**